



Finding The Smallest Or Largest Values

There may be times when you need to derive the smallest or largest value from a range, unless the smallest or largest value is zero. For instance, you might have a range of values such as {0, 3, 1, 4, 2}; in this case, the lowest value is zero, but the value you want returned is 1. There is no intrinsic function within Excel to return such a value. However, you can use a formula that will do the trick. Assuming that the range of values you want to analyze are in A4:A8, the following formula will return the lowest non-zero value:

```
=IF(MIN(A4:A8)=0,SMALL(A4:A8,COUNTIF(A4:A8,"=0")+1),MIN(A4:A8))
```

This formula uses the MIN function to determine if the lowest value in the range is zero. If it is, then the SMALL function is used to derive the lowest value, excluding the zeros. The COUNTIF function returns the number of zeros in the range, and therefore tells SMALL which item from the range to pick.

A small change to the formula allows it to be used to return the largest non-zero number in a range:

```
=IF(MAX(A4:A8)=0,LARGE(A4:A8,COUNTIF(A4:A8,"=0")+1),MAX(A4:A8))
```

These formulas will work for any range, unless the range is made up entirely of zeros. In that instance, a #NUM! error is returned.



Concatenating Cell Text

It's easy to combine the contents of your cells. Say you have a list of last names in column A, a list of first names in column B, and a list of salutations in column C. If you wanted to derive a full name for these people, you could use the following formula:

```
=C4 & " " & B4 & " " & A4
```

Smith	John	Mr	Mr John Smith
Doe	Sheila	Ms	Ms Sheila Doe
Roman	Charlie	Dr	Dr Charlie Roman
Hobbes	Larkin	Mr	Mr Larkin Hobbes

Concatenate text from different columns and display the result in another

The result is that Excel combines the values from the specified cells and places spaces between them. The ampersand character (&) is used to indicate that Excel should add text together to create a new text value.



More Useful Averages

The average value for a set is in many cases determined by removing the highest and the lowest scores and then averaging the rest. You may need to do similar types of averages. To perform this type of averaging, you can use the SMALL function. Consider the following formula, which assumes you want to find an adjusted average of the range A10:A14:

```
=(SUM(A10:A14)-SMALL(A10:A14,1)-SMALL(A10:A14,2))/(COUNT(A10:A14)-2)
```

The SMALL function is used to determine the two lowest values in the range, and these are subtracted from the overall sum of the range. The resulting value is then divided by the COUNT of values in the range. Note that the COUNT value is decreased by 2 to compensate for the fact that you are ignoring the two lowest values. Another way to calculate the same average is to use an array formula:

```
=AVERAGE(IF(A10:A14>SMALL(A10:A14,2),A10:A14))
```

Since this is an array formula, you need to enter it by pressing [Ctrl] + [Shift] + [Enter]. It examines each of the values in the range and only considers them for use in the average if they are larger than the second smallest value in the array.

While the array formula is shorter than the longer regular formula, there is one thing to remember: the array formula will produce an undesired result if there is a two-way tie in the second-lowest value in the range, or a three-way tie in the lowest value. For

instance, if the values being averaged are 3, 2, 10, 3, and 7, then the array formula will produce an average of 8.5. This is because only the values 10 and 7 are above the second-lowest value, and the average of those two is 8.5. If you use the longer formula, then the average returned is 6.666667, which is the average of 10, 3, and 7.

If these formulas don't work, check to

make sure you have the Analysis ToolPak installed. The SMALL function is a part of the Analysis ToolPak. Choose the Add-Ins option from the Tools menu and make sure there's a checkmark next to the Analysis ToolPak option.



What Is Excel Doing?

It's often frustrating to figure out exactly how Excel arrives at a particular result, particularly if the formula is complex. But Excel provides a tool you can use to help figure out what is going on when it evaluates a formula. To access this tool, select the cell containing the formula you want to evaluate. Choose Formula Auditing from the Tools menu. Excel displays a sub-menu. From the sub-menu, choose Evaluate Formula. Excel displays the Evaluate Formula dialog box.

At this point, Excel shows the full formula from the cell, and part of it is underlined. This underlined area represents the part of the formula that Excel will next evaluate. This allows you to see what intermediate steps Excel follows in arriving at a result. Every time you click the Evaluate button, Excel replaces the underlined portion of the formula with a result. Nothing you do with the formula evaluator actually affects the formula in your worksheet; it remains unchanged. Instead, Excel simply shows you what happens as it works through each part of the formula to arrive at a result.



From CAPS To Sentence Case

Some people type everything in CAPS, and if you receive a spreadsheet from one such person, what do you expect? You can use a macro to select a range of cells and convert any constants to "proper case." After you're done, the first letter of each word will be uppercase, and the rest will be lowercase. If a cell contains a formula, it is ignored. Here's the macro.

```
Sub MakeProper()  
Dim rngSrc As Range  
Dim lMax As Long, lCtr As Long  
Set rngSrc =  
ActiveSheet.Range(ActiveWindow.Selection.Address)  
lMax = rngSrc.Cells.Count  
For lCtr = 1 To lMax  
If Not rngSrc.Cells(lCtr).HasFormula  
Then  
rngSrc.Cells(lCtr) =  
Application.Proper(rngSrc.Cells(lCtr))  
End If  
Next lCtr  
End Sub
```

The "l" in "lCtr" and "lMax" is a lowercase "L." ■